4. Sequence Diagram

Online Exam System: MFA, Autosave, Anti-Cheat, Grading & Notifications
Student (Desktop)
Web Client (HTML/CSS)
Express Gateway (+ middleware)
Auth Module
Exam Module
Grading Module
WebSocket Hub
MySQL
Redis (optional)
SendGrid
Teacher
Admin
rate limit 100//min, CSRF, XSS filters
Enter credentials (+ MFA if enabled)
POST /auth/login (HTTPS)
Forward
SELECT user; verify bcrypt hash
user record
Alternative
[[MFA enabled]]
Verify MFA TOTP
Store refresh token jti
OK
{ access_jwt (15–30m), refresh_jwt (1–3d) }
{ access_jwt, refresh_jwt }
Login success
Open exam
GET /exam/{id} (Bearer access_jwt)
Forward
Validate JWT + revocation list
OK (sub, role=Student)
Load IMMUTABLE snapshot + apply per-student seed
INSERT exam_session (state=Active, expires_at)
session_id
Questions randomized (options shuffled, pool subset)
Exam payload + session_id
Exam payload + session_id
Render exam + timer
Open WSS /session/{id} (subprotocol: jwt)
Validate JWT
OK
Connected; push timer ticks
Loop
[[While answering]]
Answer question
autosave(answer, qid)
Persist draft
UPDATE answers
OK
OK
ack
Anti-cheat events reported on same WSS channel
cheat_event(type, ts, fingerprint)
Record event
INSERT cheat_event
OK
OK
ack
Alternative
[[Connection drops]]
disconnect
Reconnect with JWT + session_id
Validate session not expired (server-side)
SELECT exam_session by session_id
session active?
Alternative
[[Expired]]
Reject; session expired
session expired
[[Active]]
Resume state
Resume + latest state
Alternative
[[Reload (within window)]]
Reload
GET /exam/{id} (Bearer access_jwt)
Forward
Validate JWT
OK
GET exam payload + current draft
payload + draft
payload + session_id
payload + session_id
(re)Open WSS /session/{id}
Connected; push timer ticks
Alternative
[[Click Submit]]
Click Submit
POST /exam/{id}/submit
Forward
Validate JWT
OK
Hard cutoff check
Mark submission Final
OK
Grade(submission_id)
Load answer key + cheat events
key + events
Auto-grade MCQs; apply weights & negatives
Save partial grade (status=AwaitingReview if open Qs)
OK
Append audit log (hash-chained)
OK
Queue notification email
queued
Result
Submission receipt + provisional result
Receipt + provisional result
Show provisional result
[[Server timer expires]]
Hard cutoff -> auto-submit
Mark submission Final
OK
Grade(submission_id)
Load answer key + cheat events
key + events
Auto-grade MCQs; apply weights & negatives
Save partial grade (status=AwaitingReview if open Qs)
OK
Append audit log (hash-chained)
OK
Queue notification email
queued
Result
Receipt + provisional result
Show provisional result
Review open questions / override grade
Forward
Update final grade + feedback
OK
Append audit log (hash-chained)
OK
Notify student (grade updated)
queued
Manage users/exams, enable/disable MFA, view audit logs
Update MFA settings / roles
UPDATE user settings/roles
OK
Append audit log (hash-chained)
OK
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